

RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF TECHNOLOGY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 3219 – Mobile Application Development

Practical Sheet – 01

Understand the purpose and use cases of different Android layouts

OBJECTIVES

1. Understand the purpose and use cases of different Android layouts: Frame Layout, Linear Layout, Constraint Layout, Relative Layout, Grid View, and Table Layout.
2. Design visually appealing user interfaces using XML in Android Studio.
3. Implement layout hierarchies by nesting layouts to build structured UI designs.
4. Use and style common UI widgets such as Text View, Button, Edit Text, and Image View within different layouts.
5. Apply layout-specific attributes like gravity, weight, constraints, alignment rules, and spacing.
6. Create real-world user interface designs such as login screens, profile cards, and photo galleries.

Task 01 – Frame Layout

Step 1: Open Android Studio

1. Launch **Android Studio** from your computer.
2. Click on **"New Project"** or open an **existing project**

Step 2: Create a New Activity (Optional if you already have one)

1. In the **Project view**, right-click on Kotlin+java > your package name.
2. Go to **New > Activity > Empty Activity**.
3. Name it (e.g., frame_layout) and click **Finish**.

Step 3: Open frame_layout.xml

1. Go to res > layout > frame_layout.xml.
2. Switch to **Code view** (not Design view).
3. **Replace the existing code** with the provided Frame Layout code.

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Background Image -->
    <ImageView
        android:contentDescription="@null"
```

```
        android:id="@+id/backgroundImage"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:scaleType="centerCrop"
        android:src="@drawable/background" />

    <View
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:background="#66000000" />

    <!-- CardView -->
    <androidx.cardview.widget.CardView
        android:layout_width="350dp"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:padding="20dp"
        app:cardCornerRadius="10dp"
        app:cardElevation="10dp"
        android:background="@android:color/white">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:gravity="center"
            android:padding="20dp">

            <!-- Logo -->
            <ImageView
                android:contentDescription="@null"
                android:layout_width="200dp"
                android:layout_height="200dp"
                android:src="@drawable/logo1"
                android:layout_marginBottom="12dp" />

            <!-- Welcome -->
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="@string/welcome"
                android:textSize="24sp"
                android:textColor="@color/white"
                android:textStyle="bold"
                android:layout_marginBottom="20dp" />

            <!-- Get Started Button -->
            <Button
                android:id="@+id/btnGetStarted"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="@string/get_started"
                android:textColor="#FFFFFF"
                android:backgroundTint="#3F51B5"
                android:textSize="18sp"
                android:padding="10dp"
                android:layout_marginTop="10dp" />

        </LinearLayout>

    </androidx.cardview.widget.CardView>

</FrameLayout>
```

4. Make sure that, app > manifests >AndroidManifest.xml contain this code.

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:label="@string/app_name"
            android:theme="@style/Theme.AppCompat.Light.NoActionBar"
            android:windowSoftInputMode="adjustResize">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

5. Make sure that, app > values >themes.xml contain this code.

```
<?xml version="1.0" encoding="utf-8"?>
<resources>

    <style name="Theme.MainActivity" parent="Theme.AppCompat.Light.NoActionBar" />
</resources>
```

6. Make sure to add the string values to app > values >string.xml contain this code.

```
<resources>
    <string name="app_name">Main Activity</string>
    <string name="welcome">Welcome</string>
    <string name="get_started">Get Started</string>
</resources>
```

Step 4: Add CardView Dependency

1. Open build.gradle (Module: app).
2. Add this inside dependencies { ... }:
3. implementation("androidx.cardview:cardview:1.0.0")
4. Click "**Sync Now**" when prompted.

Step 5: Add the background and logo1 image to Drawable Folder

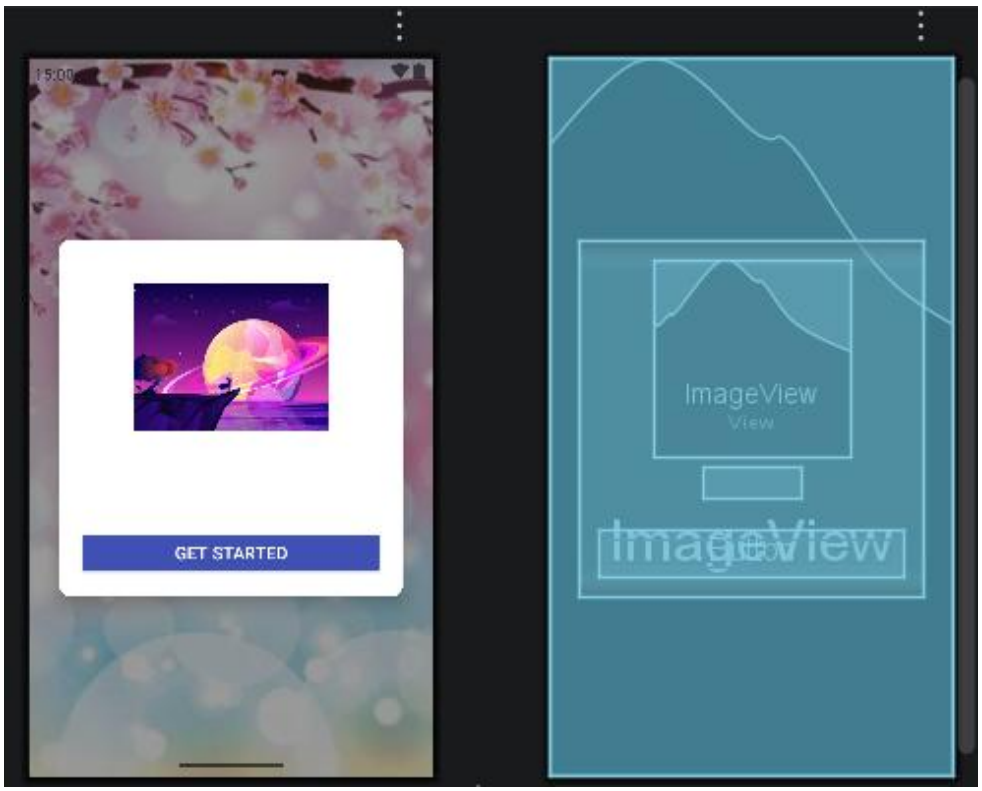
1. Copy your background image and logo1 image (e.g., background.jpg, logo1.jpg) into the res/drawable folder.
2. Rename it to something like background.jpg.
3. Make sure to keep the XML with the same name of the image

    xml
    CopyEdit
    android:src="@drawable/background"

android:src="@drawable/logo1"

Step 6: Connect a virtual Device

1. Go to the Device Manager look for a virtual device if not by clicking the plus icon create virtual device.
2. Select the suitable device something like pixel 6.
3. Download the device.
4. By clicking the finish button add the device to android studio.
5. Once added the device click the play button to start the virtual device.
6. Finally, by clicking the run launch the app in the device.



Activity 01

- There is a simple mistake in the designing part, find out and redesign?

Step 7: Create a New Layout XML File

For each layout (Linear, Constraint, etc.), you'll create a separate XML file.

1. Right-click on res > layout.
2. Click **New > Layout Resource File**.
3. Name the file, e.g., activity_linear.xml.
4. Choose Root Element: LinearLayout, ConstraintLayout, etc., based on layout type.
5. Click **OK**.

Task 02 – Linear Layout

Step 1: Open linear_layout.xml

1. Go to res > layout > linear_layout.xml.
2. Switch to **Code view** (not Design view).
3. **Replace the existing code** with the provided Linear Layout code.

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="#d6d5e3"
    android:gravity="center">

    <ImageView
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@drawable/profilepicture"
        android:layout_gravity="center"
        android:scaleType="centerCrop"
        android:background="@drawable/circle_bg"
        android:contentDescription="@string/profile_picture_description"
        android:clipToOutline="true" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Jane Doe"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#37474F"
        android:layout_marginTop="16dp" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Android Developer"
        android:textStyle="bold"
        android:textColor="#607D8B"
        android:layout_marginTop="4dp" />

</LinearLayout>
```

Step 2: Create circle_bg.xml

1. Right-click res > drawable
2. Click New > Drawable Resource File
3. File name: circle_bg
4. Root element: shape ← type this
5. Click OK
6. Replace everything inside with this code:

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">

    <size android:width="120dp" android:height="120dp"/>

</shape>
```

Step 3: Find the MainActivity

1. In left panel, go to kotlin+java
2. Open com.example.mainactivity
3. Open MainActivity.kt
4. Look for setContentView(R.layout.___)
5. Change the layout name to linear_layout

Step 4: Launch the app

6. Select the suitable device and click the play button to start the device.
7. Now run and launch the app in the device.



Task 03 – Constraint Layout

Step 1: Open constraint_layout.xml

1. Go to res > layout > constraint_layout.xml.
2. Switch to **Code view** (not Design view).
3. **Replace the existing code** with the provided Linear Layout code.

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    android:background="#FAFAFA">

    <EditText
        android:id="@+id/username"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="80dp"
        android:backgroundTint="#3F51B5"
        android:hint="Username"
        android:minHeight="48dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/password"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:backgroundTint="#3F51B5"
        android:hint="Password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/username"
        android:importantForAutofill="no" />

    <Button
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="Login"
        app:layout_constraintTop_toBottomOf="@id/password"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="24dp"
        android:backgroundTint="#3F51B5"
        android:textColor="#FFF" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Step 2: Find the MainActivity

1. In left panel, go to kotlin+java
2. Open com.example.mainactivity
3. Open MainActivity.kt
4. Look for setContentView(R.layout.___)
5. Change the layout name to constraint_layout

Step 3: Launch the app

1. Select the suitable device and click the play button to start the device.
2. Now run and launch the app in the device.



Activity 02

Add Following controls to your Constraint Layout

- "Remember Me" check box
- "Forgot Password?" clickable text view
- A Register prompt (to guide users to create an account)
- Change background color

```
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    android:background="#E8F5E9">

    <EditText
        android:id="@+id/username"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="80dp"
        android:backgroundTint="#388E3C"
        android:hint="Username"
        android:minHeight="48dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/password"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:backgroundTint="#388E3C"
        android:hint="Password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        android:importantForAutofill="no"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/username" />

    <!-- Remember Me CheckBox -->
    <CheckBox
        android:id="@+id/rememberMe"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Remember Me"
        android:textColor="#388E3C"
        android:layout_marginTop="12dp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/password" />

    <!-- Forgot Password TextView -->
    <TextView
        android:id="@+id/forgotPassword"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Forgot Password?"
        android:textColor="#388E3C"
        android:textStyle="bold"
        android:clickable="true"
        android:focusable="true"
```

```
        android:layout_marginTop="12dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@id/password" />

    <!-- Login Button -->
    <Button
        android:id="@+id/loginButton"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="24dp"
        android:backgroundTint="#388E3C"
        android:textColor="#FFF"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/rememberMe" />

    <!-- Register Prompt -->
    <TextView
        android:id="@+id/registerPrompt"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Don't have an account?"
        android:textColor="#555555"
        android:layout_marginTop="16dp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/loginButton" />

    <TextView
        android:id="@+id/registerLink"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Register Here"
        android:textColor="#388E3C"
        android:textStyle="bold"
        android:clickable="true"
        android:focusable="true"
        app:layout_constraintStart_toEndOf="@id/registerPrompt"
        app:layout_constraintTop_toBottomOf="@id/loginButton"
        android:layout_marginTop="16dp"
        android:layout_marginStart="4dp" />

</androidx.constraintlayout.widget.ConstraintLayout>
```